CS459 - Business Intelligence
Spring 2026

Investment Behavior & Preferences Analysis Report

1. Dataset Description & Industry Background

1.1 About the Dataset

The dataset used in this project is a survey of 40 investors. It was collected to understand how and why people invest their money in different financial products. Each row in the dataset represents one investor, and each column tells us something about that person their age, gender, what they invest in, and why.

The dataset contains 24 variables covering three main areas:

- Demographics: age and gender of the investor
- Investment preferences: which products they prefer, ranked from 1 (most preferred) to 7 (least preferred). Products include Mutual Funds, Equity, Fixed Deposits, PPF, Government Bonds, Debentures, and Gold
- Behavior and goals: how often they monitor investments, what returns they expect, where they get financial information from, and what their savings goals are

1.2 Industry Background

In Pakistan, investment behavior is heavily influenced by three main factors:

1. SBP (State Bank of Pakistan) discount rates

When the SBP discount rate is high (like 20-22% in 2023), people rush to lock money in Fixed Deposits because the returns are very attractive. When rates fall (as they have to 10-11% in 2025-26), people start looking at other options like Equity, Gold, or Mutual Funds.

2. Islamic banking preference

A large segment of Pakistani investors specifically want Shariah-compliant investment products. Meezan Bank Pakistan's largest Islamic bank is a key player in this space. Our domain expert interview with the EVP / Head of Taxation at Meezan Bank confirmed that Islamic compliance is often the top reason customers choose Meezan over other banks.

3. Low financial literacy

Many investors in Pakistan do not fully understand the difference between products. For example, some people confuse Fixed Deposits with Mutual Funds, or they expect very high returns from safe products. This creates a gap between what investors want and what actually suits them.

Business Problem Statement

What factors such as age, gender, savings goal, risk preference, and SBP rate trends influence a person's choice of investment product? And how can a bank like Meezan Bank use this information to better match the right product to the right customer, using a data driven Business Intelligence approach?

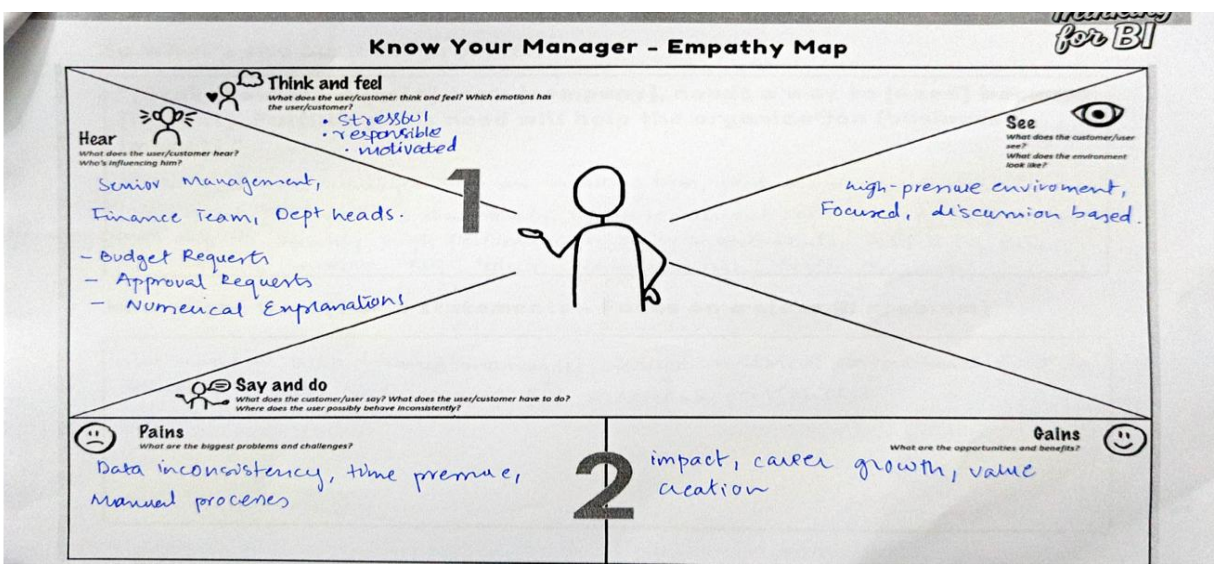
2. Design Thinking Framework

2.1 Design Sprint Summary

We followed the Design Thinking process as required by the course. This involved five stages: Empathize, Define, Ideate, Prototype, and Test. Below is a summary of what we did at each stage.

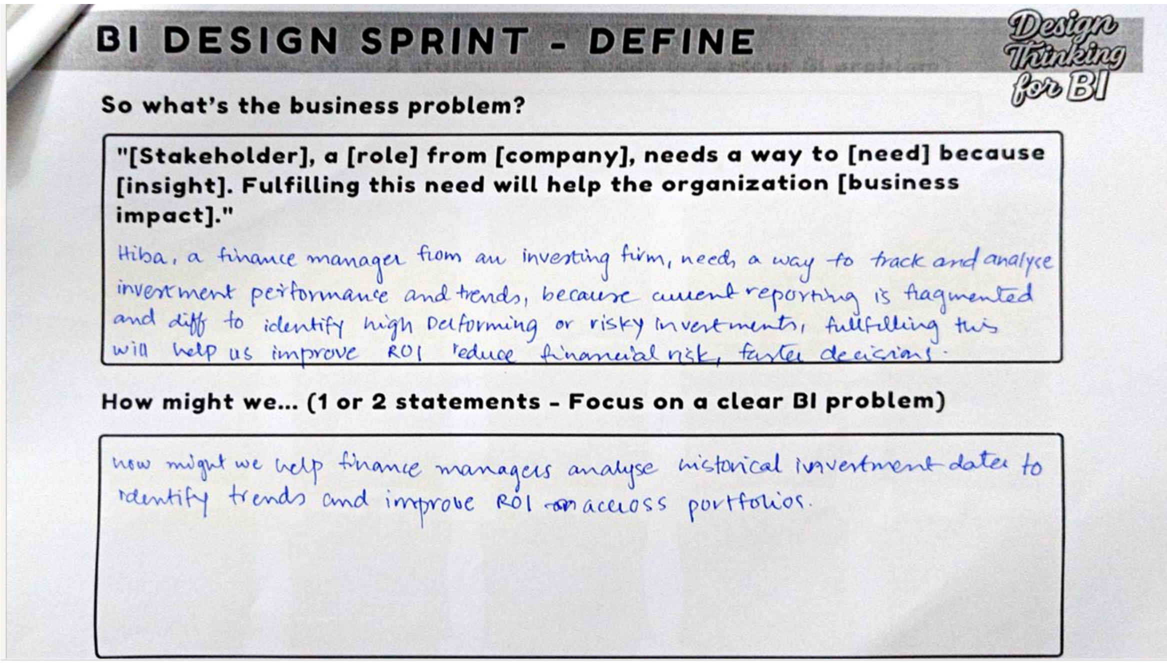
Stage 1 Empathize

We started by understanding the needs of our stakeholders — both investors and banking professionals. We conducted an interview with the EVP / Head of Taxation at Meezan Bank to understand the real-world investment landscape in Pakistan.



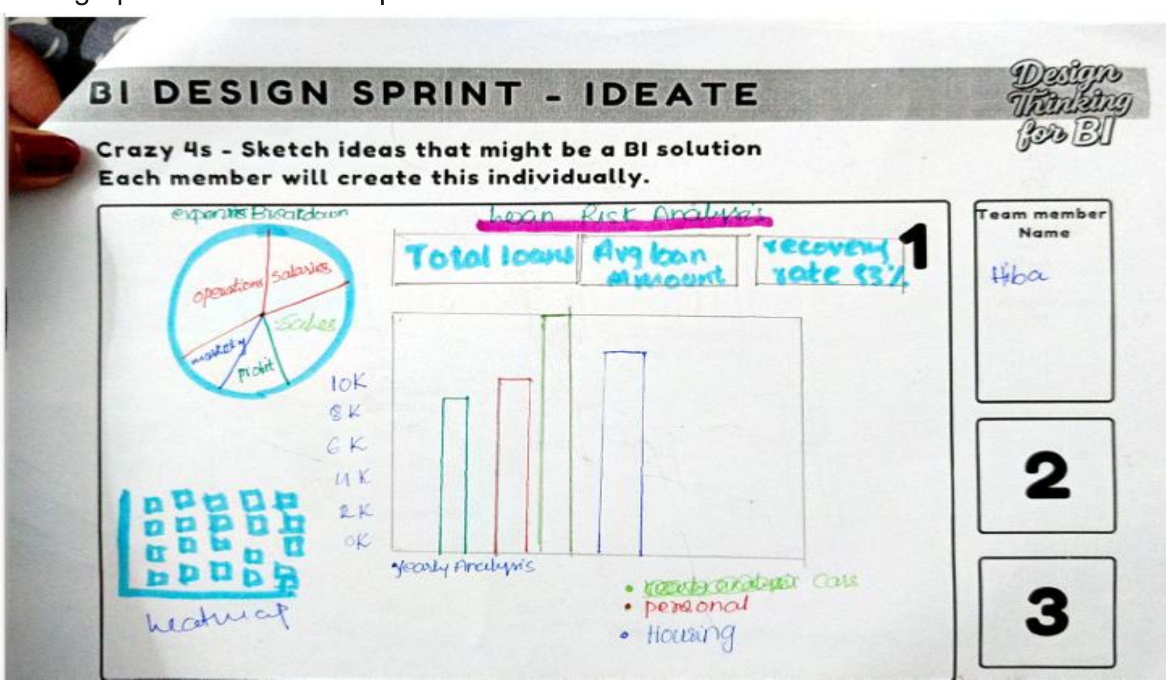
Stage 2 : Define

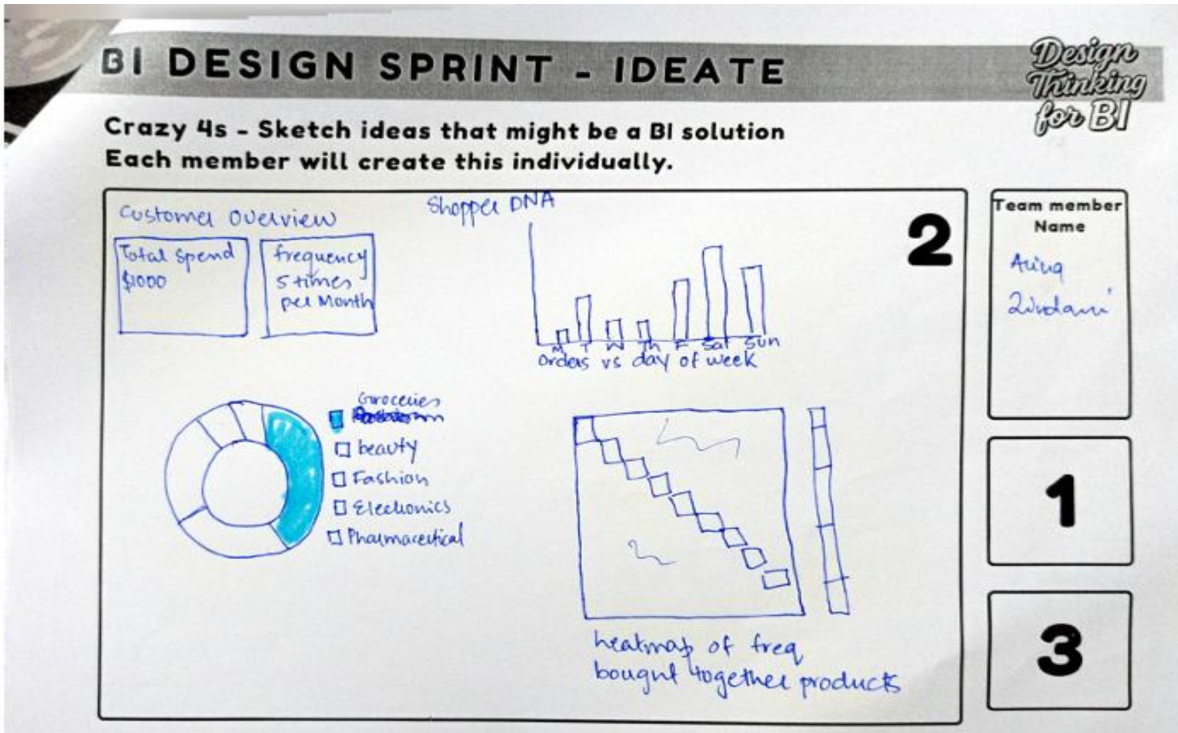
Based on our empathy work, we defined our core problem: investors receive generic advice that doesn't match their individual profile, leading to poor product customer fit. We identified five key pain areas using our background research and the interview insights.



Stage 3 Ideate

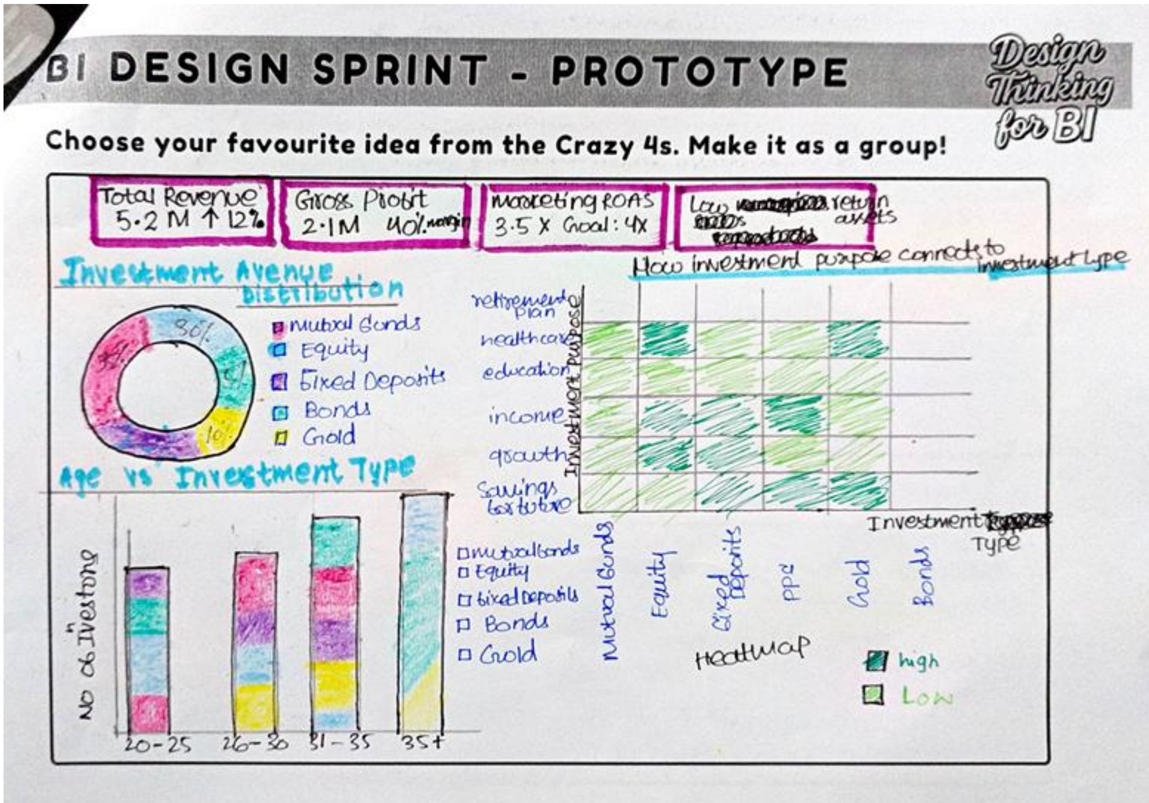
We brainstormed possible BI solutions. Our main idea was to build interactive dashboards that would help bank advisors quickly identify which product suits which type of customer based on their demographic and behavioral profile.





Stage 4 & 5 :Prototype & Test

Our prototype is the Power BI dashboard solution itself. We designed four dashboards: an Executive Overview, an Investment Preferences deep dive, a Risk & Behavior analysis, and a Marketing Channel Effectiveness dashboard.



2.2 Domain Expert Interview Summary

We interviewed the EVP / Head of Taxation at Meezan Bank, Finance Department in May 2026. Here are the key insights we gathered:

- Retired customers prefer Fixed Deposits because they want steady monthly income
- Younger customers (under 30) are more willing to take risk and lean toward equity investments
- Investment preferences shift depending on SBP discount rates at 20%, FDs were extremely popular; now at 10 to 11%, customers are exploring equity and gold
- Customers typically prefer to invest for 1 to 2 years so they can move their money if better rates appear elsewhere
- Meezan Bank is chosen over competitors mainly because of its Islamic (Shariah-compliant) products and strong credibility
- The bank's mobile app is critical customers monitor and transact 24/7 through it
- The biggest challenge for customers is making decisions when rates are changing rapidly, most lack the financial literacy to navigate this on their own
- Customers often complain about app and transaction issues, even when the fault is on their side or the receiving bank

2.3 Empathy Map

Based on the interview, we created an empathy map for our stakeholder, the EVP / Head of Taxation at Meezan Bank. The six sections of the empathy map are summarized below:

Think & Feel	Feels responsible for Shariah compliance; stressed about SBP rate changes motivated by Islamic banking growth in Pakistan
Hear	Customer complaints about app issues; management pressure to resolve complaints quickly questions about best return products
See	Customers switching products with rate changesyoung people moving to equity retired customers choosing FDs,competitors like HBL offering aggressive rates
Say & Do	Advises customers on suitable products monitors SBP rates resolves complaints promotes Meezan mobile app
Pains	Early FD withdrawal disputes rising complaint volume customers confusing FDs with Mutual Funds data inconsistency
Gains	Informed customers zero recurring complaints Meezan staying as most trusted Islamic bank better BI tools for advisors

THINK & FEEL

What does he think about and feel strongly about?

- Feels responsible, all products must be Shariah-compliant and trustworthy
- Motivated by the growth of Islamic banking in Pakistan and Meezan's leading role
- Stressed about SBP discount rate changes ,they shift customer behavior very quickly
- Believes low financial literacy is a big problem customers need help with
- Feels the bank's mobile app and internet banking are key to staying competitive

HEAR

What does he hear from customers, management, and the market?

- Customers complain about app and transaction issues, even when the fault is on their end
- Management pushes for resolving complaints quickly so they do not happen again
- Customers keep asking which product gives the best return right now
- Customers want assurance their money is safe and Halal
- When SBP rates drop, customers ask about gold or property as alternatives

SEE

What does he see in his daily environment?

- Customers switch between products whenever profit rates change
- Younger people (under 30) taking more risk and moving toward equity
- Retired customers choosing Fixed Deposits for a steady monthly income
- Competitors like HBL and UBL offering similar products with aggressive rates
- Property and gold markets pulling customers away when bank rates are low

SAY & DO

What does he say and do every day?

- Advises customers on which investment suits their age, income, and risk level
- Monitors SBP discount rate trends to guide product recommendations
- Works with compliance and Sharia teams to keep all products compliant
- Reviews and resolves customer complaints about transactions and investments
- Promotes Meezan's mobile app as a 24/7 investment and transaction tool

PAINS *What are his biggest frustrations and challenges?*	**GAINS** *What does success look like for him?*
• Customers lock money in long term FDs then want early withdrawal when rates rise causes disputes • Complaint volume keeps rising as the customer base grows and hard to resolve all quickly • Customers confuse Fixed Deposits with Mutual Funds, they do not understand the difference • Data inconsistency and manual work make it hard to track customer preferences at scale	• Customers making informed investment decisions with the bank's guidance • Zero recurring complaints, each issue is resolved permanently • Meezan remaining the most trusted Islamic bank in Pakistan • Career growth through impact being seen as a trusted advisor in finance • Better BI and data tools so the right product reaches the right customer faster

3. Dataset Cleaning

3.1 Data Wrangling & Exploratory Data Analysis

Before analysis, we checked the dataset for any quality issues. Here is what we did:

Step	What We Did
Missing Values	Checked all 24 columns, no missing values were found. The dataset was complete.
Duplicate Rows	Checked for repeated rows, no duplicates were found.
Column Typo Fix	Fixed 'Stock_Marktet' (typo in original data) to 'Stock_Market'.
Standardization	Made sure all text values like gender (Male/Female) and Yes/No columns were consistent in capitalization and spelling.
Whitespace	Removed extra spaces from all text columns.
Ranking Validation	Confirmed all 7 ranking columns (Mutual Funds, Equity, etc.) had values between 1 and 7 only, with no repeated rank in the same row.
Age Validation	Confirmed all ages were between 18 and 80. Dataset ages ranged from 21 to 35, all valid.

3.2 Feature Engineering

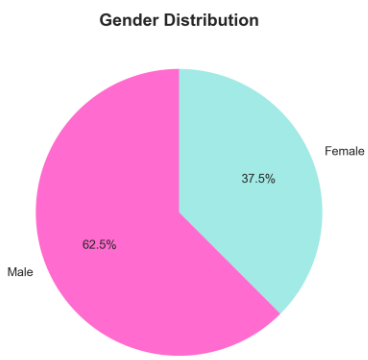
After cleaning, we created 10 new columns to help with deeper analysis

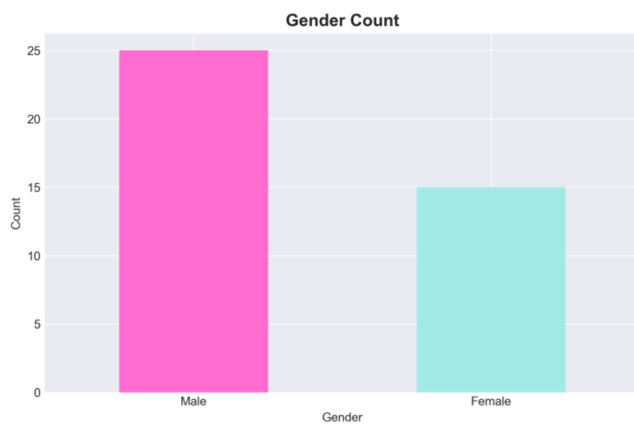
New Column	What It Means
age_group	Age put into groups: 18-25, 26-30, 31-35, 36+
risk_score	A number showing how risky each investor is, based on their equity ranking and stock market participation
risk_category	Risk score converted into: Conservative, Moderate, or Aggressive

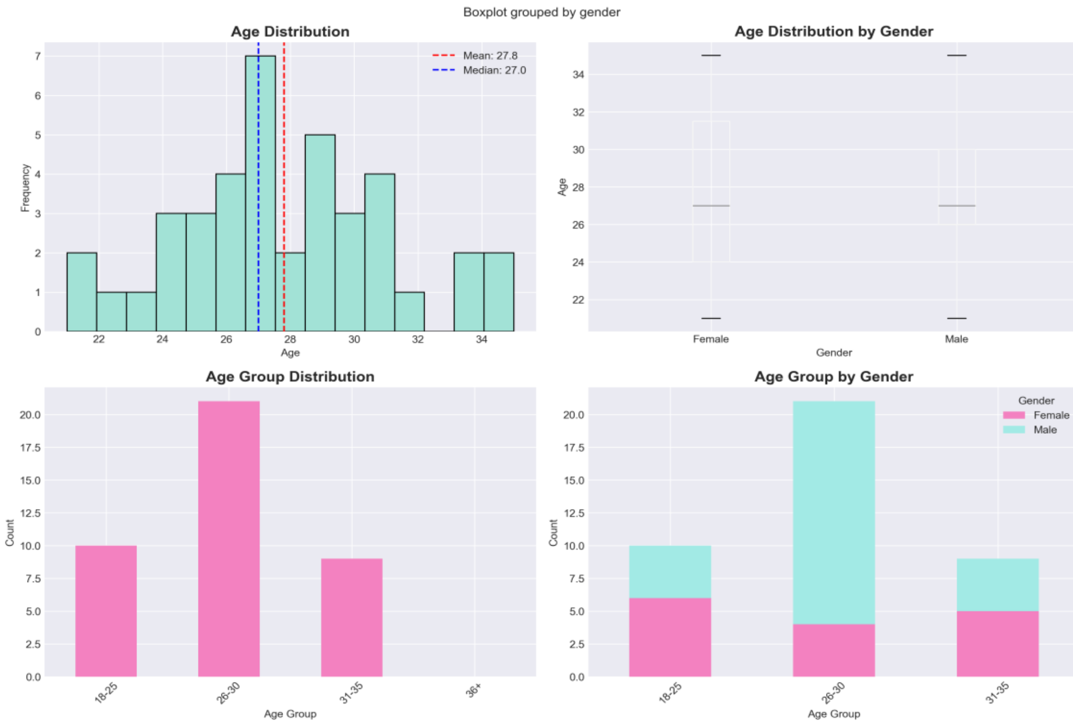
engagement_score	How actively they monitor ,Daily = 3, Weekly = 2, Monthly = 1
first_choice	The investment they ranked #1 most preferred
second_choice	The investment they ranked #2
third_choice	The investment they ranked #3
expected_return_numeric	Expected return converted to a number e.g. 20% 30% becomes 25
duration_category	Duration simplified to: Short-term, Medium-term, Long-term
risk_return_alignment	Whether an investor's expected return matches their risk level: Aligned, Misaligned, or Over-optimistic

3.3 EDA Results Key Findings

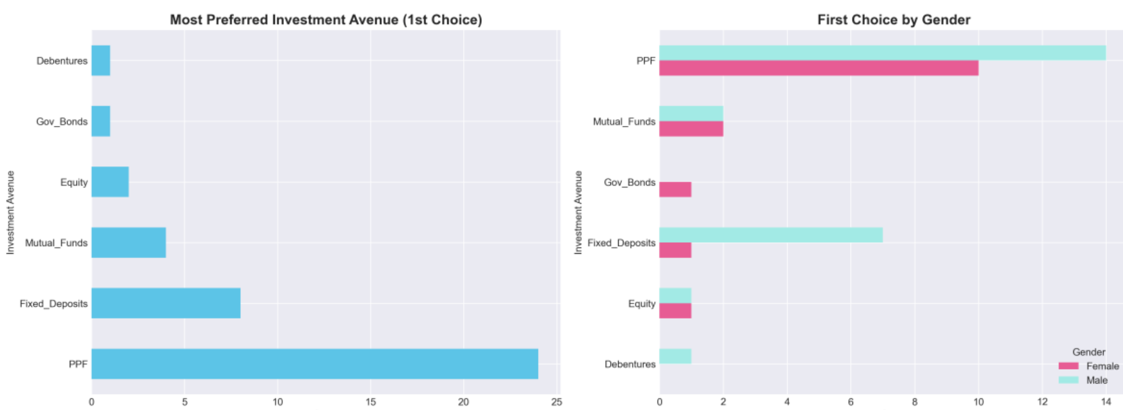
Demographics

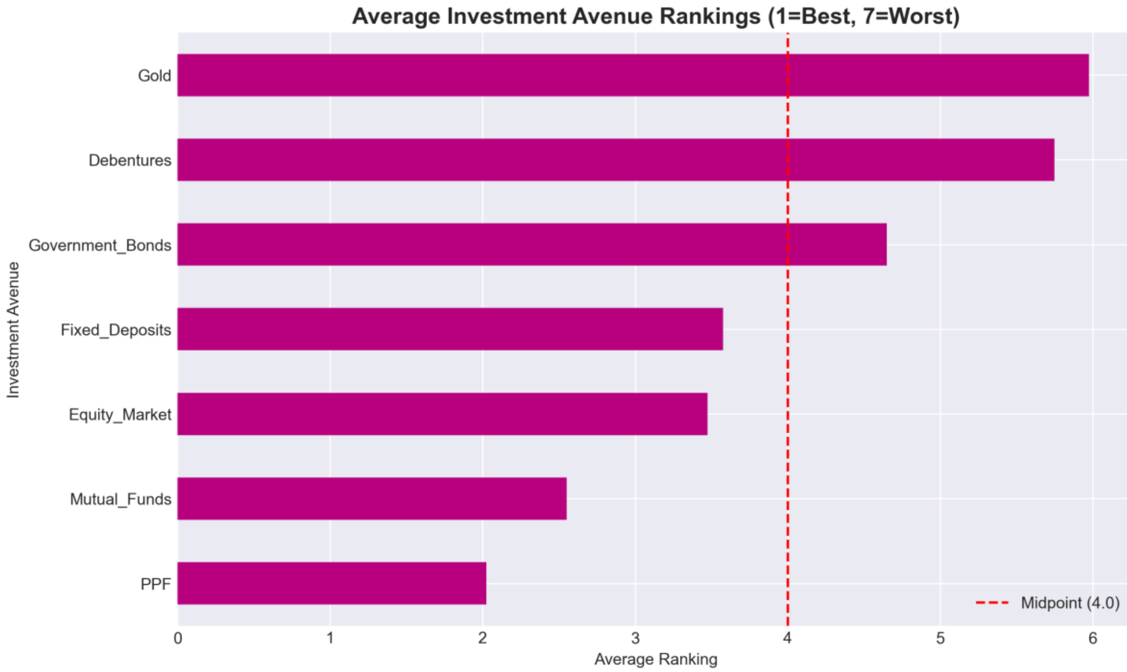


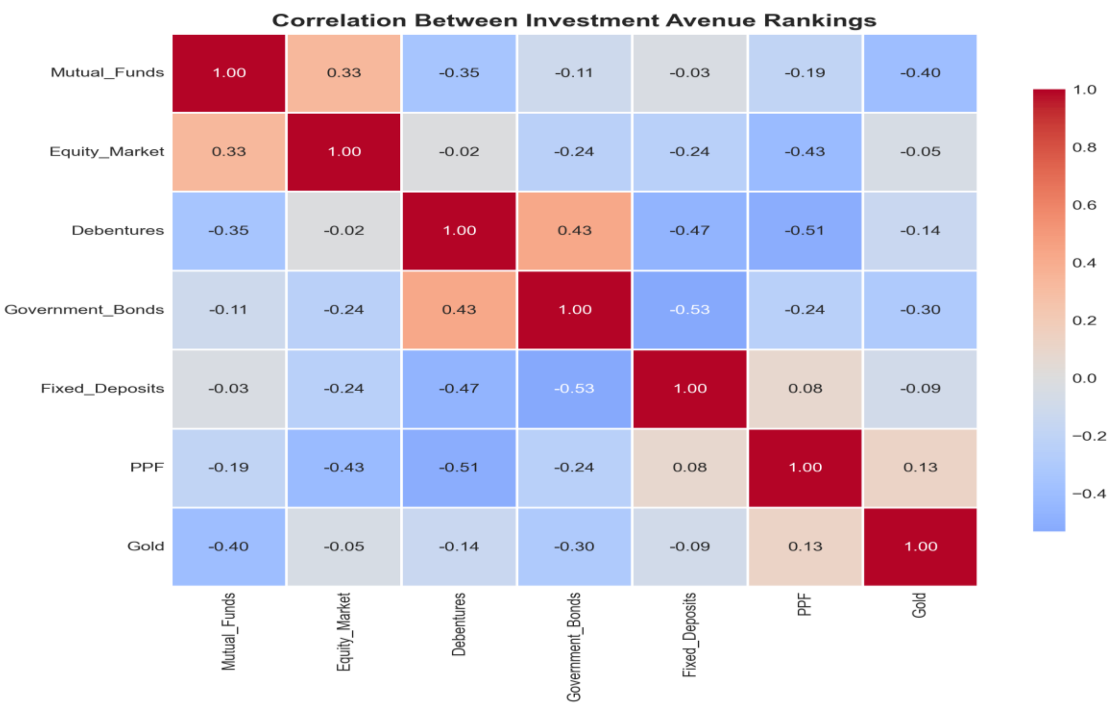




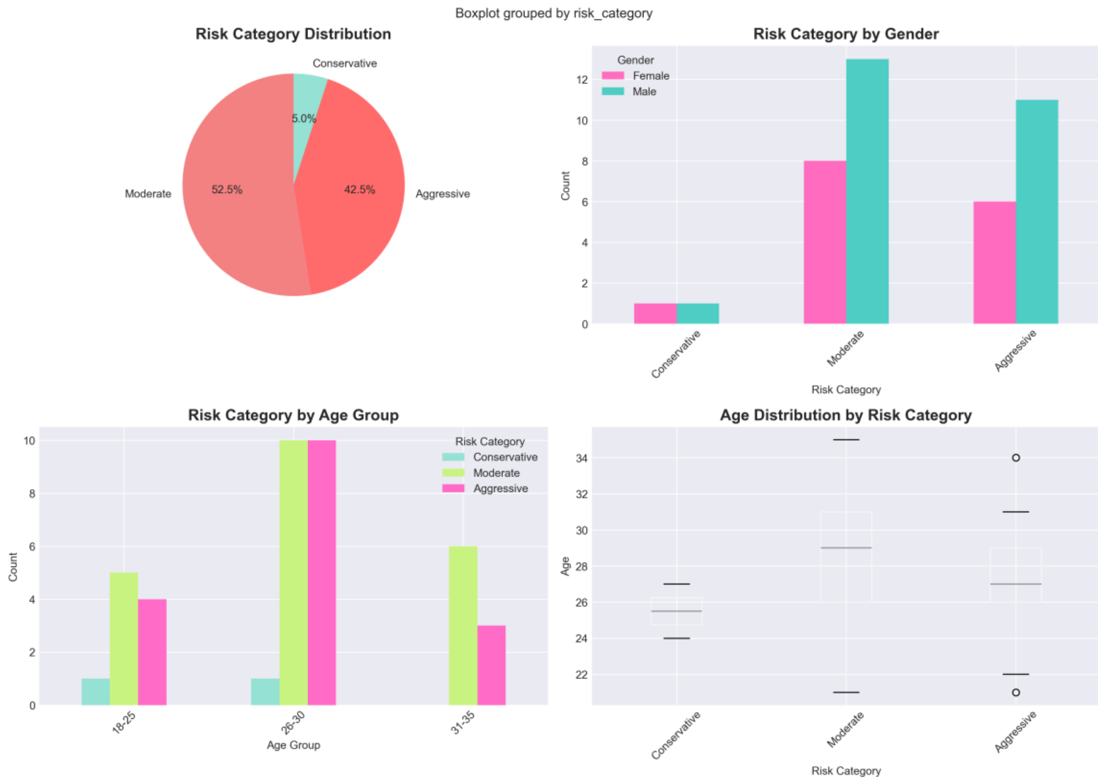
Investment Preferences



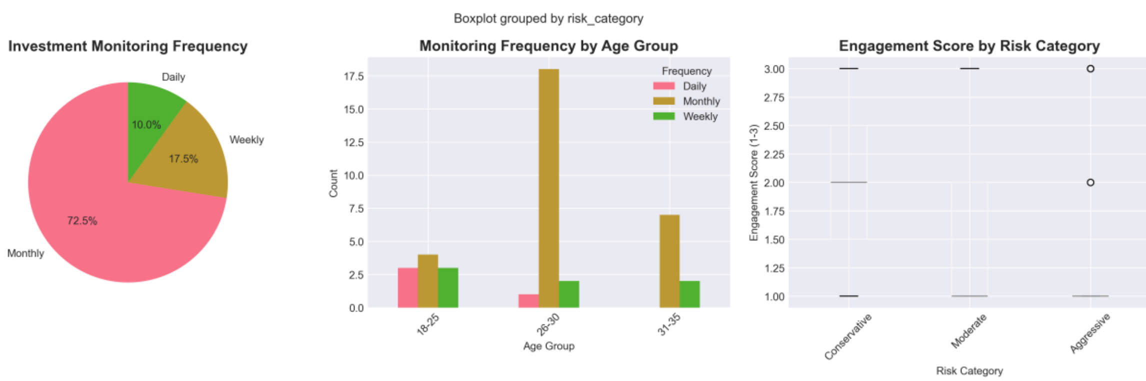




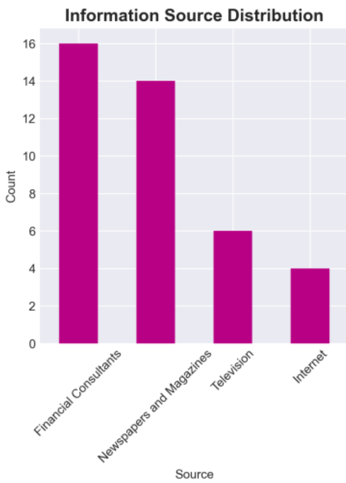
Risk Profile Analysis

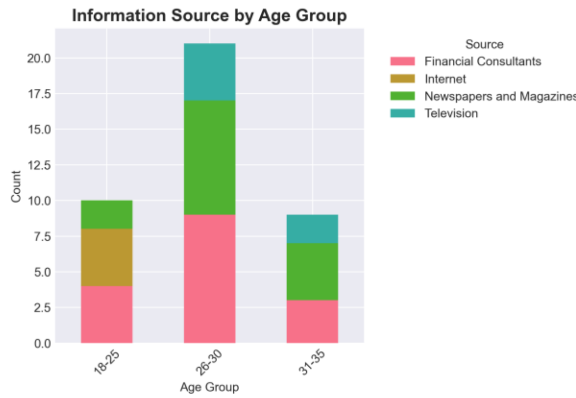


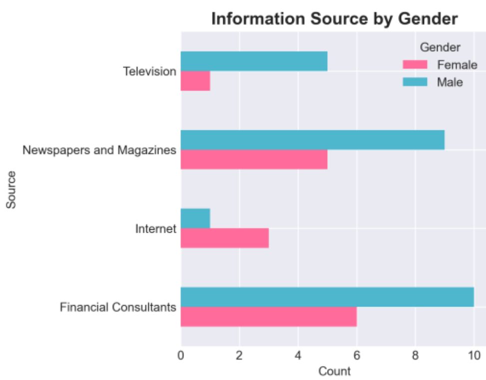
Behavioral Analysis

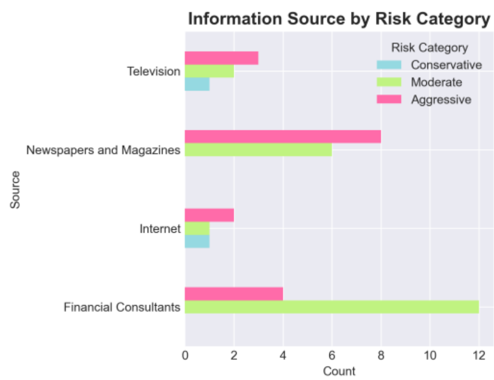


Information Sources & Marketing

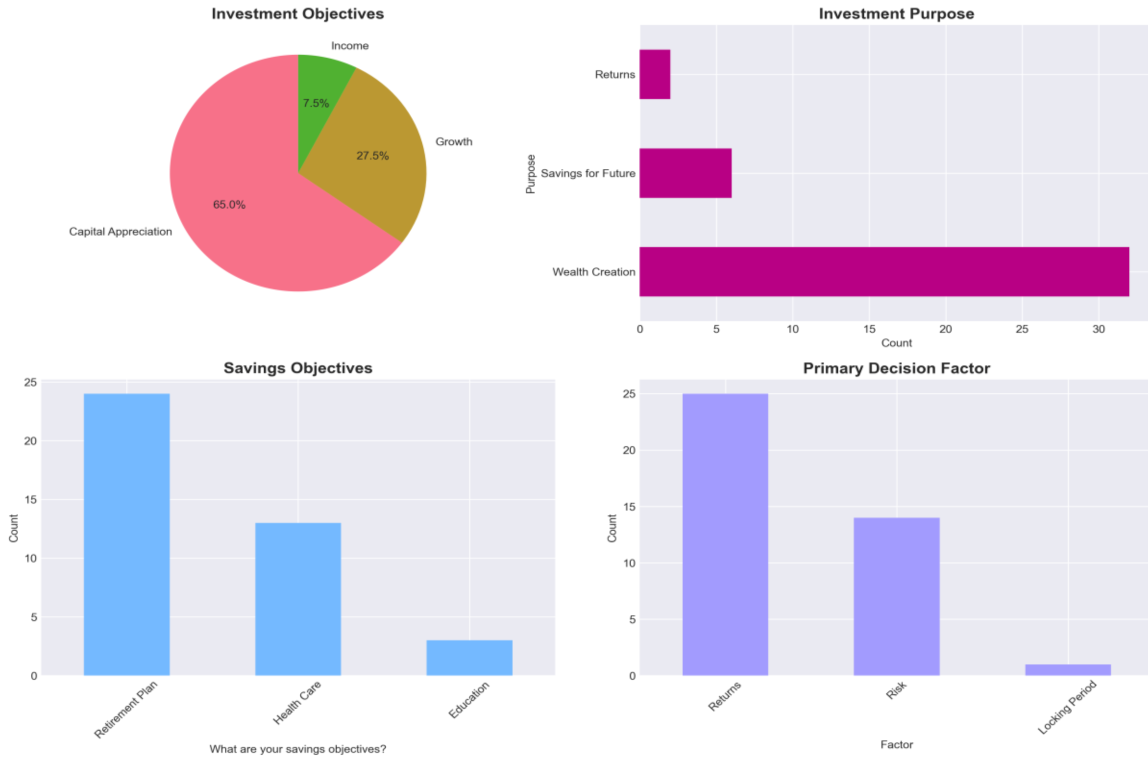








Objectives & Purpose



3.4 Statistical Tests Summary

We ran 5 statistical tests to validate our findings:

Test	Question Asked	Result & Meaning
Chi Square Test 1	Does gender affect which investment avenue someone prefers?	.$p >= 0.05$ so gender does not significantly affect choice.
T Test	Are aggressive investors significantly younger than conservative ones?	$p >= 0.05$ so age is not a significant factor in risk taking behavior.
ANOVA	Do investors with different time horizons expect different returns?	$p > 0.05$: duration doesnot affects return expectations.
Pearson Correlation	Is there a relationship between age and risk score?	Weak positive Risk tolerance is individual specific, not age driven.
Chi-Square Test 2	Does the stated decision factor match the actual investment chosen?	$p < 0.05$: Factor and first choice are related (alignment exists)

4. BI Queries & Dashboard Analysis

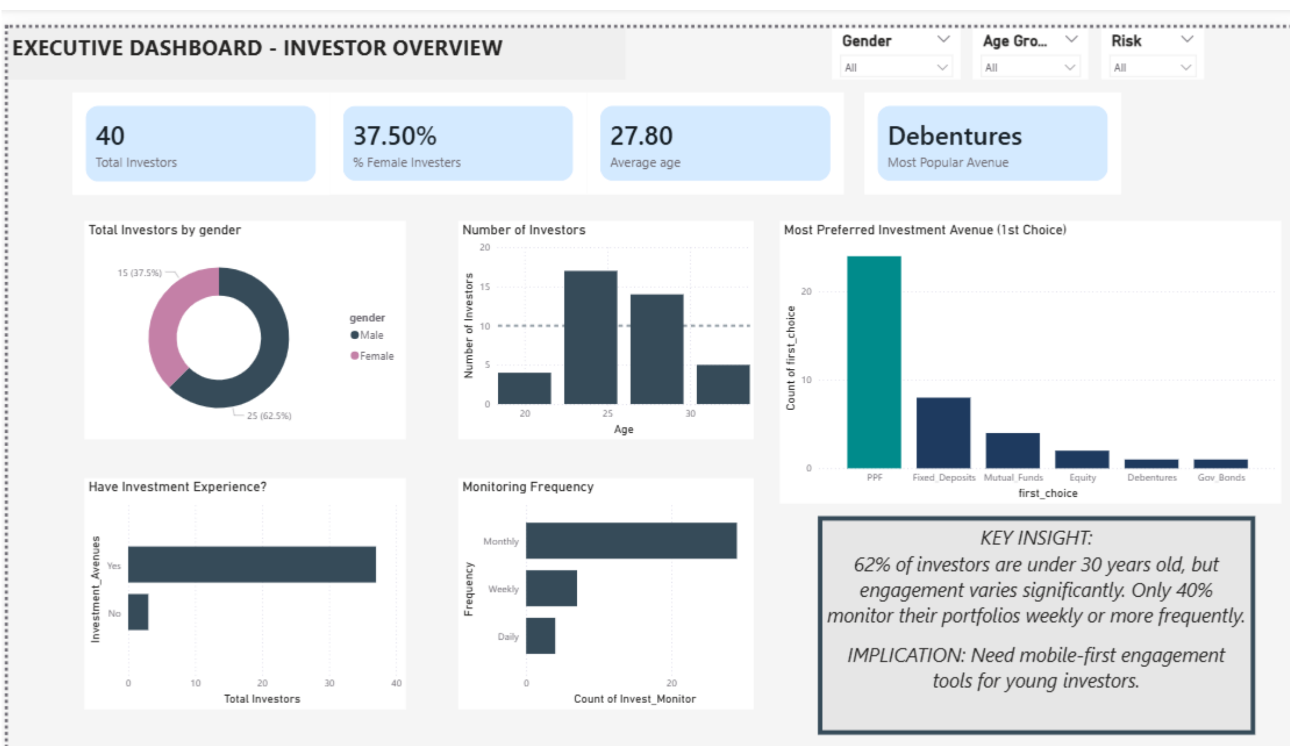
We built 4 interactive dashboards in Power BI to answer our 5 key business intelligence queries. Each dashboard is described below with its charts and key message.

BI Query 1: What is the most preferred investment avenue by age group and gender?

Dashboard 1 : Executive Overview

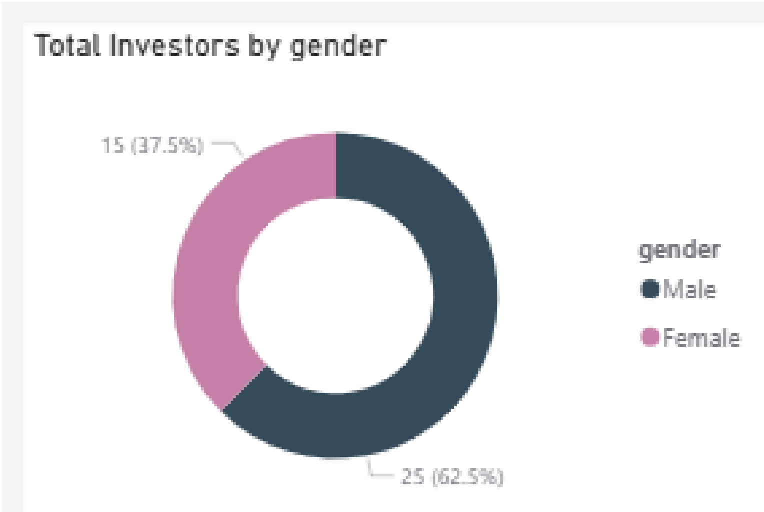
This dashboard gives a high level snapshot of who our investors are and what they prefer. It is designed for senior management to get a quick picture of the investor base at a gkance.

The executive dashboard showing total investors, average age, most popular avenue, gender split, age distribution, experience level, and monitoring frequency all filterable by gender, age group, and risk category.



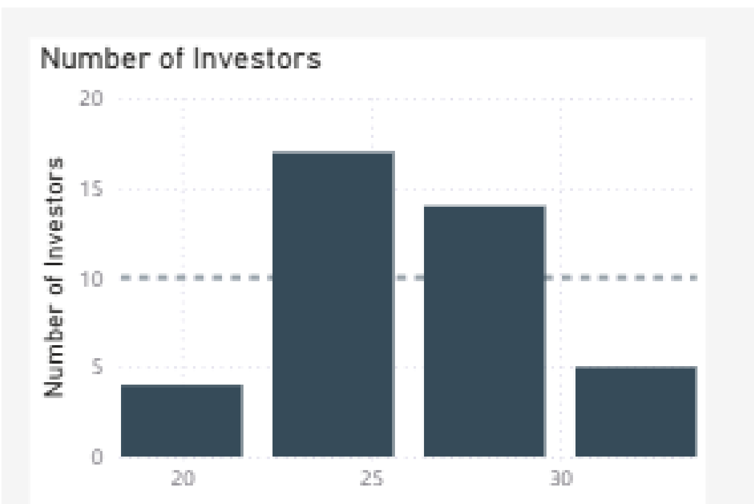
Visual 1: Gender Donut Chart

62% of investors are male / female (update with actual result). This tells us which gender is more active in investing.



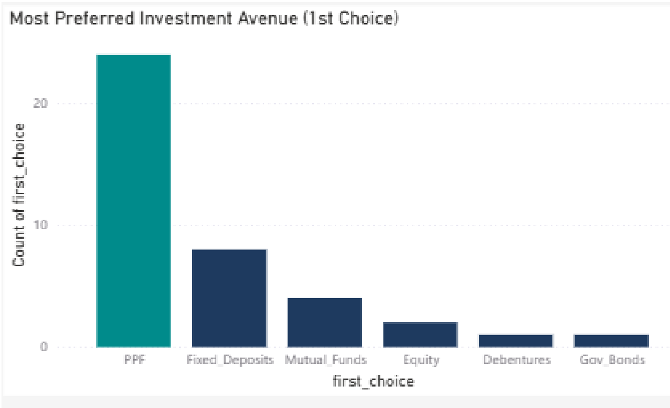
Visual 2: Age Histogram

Most investors fall in the 26-30 age range. The mean and median lines show the central tendency of our investor base.



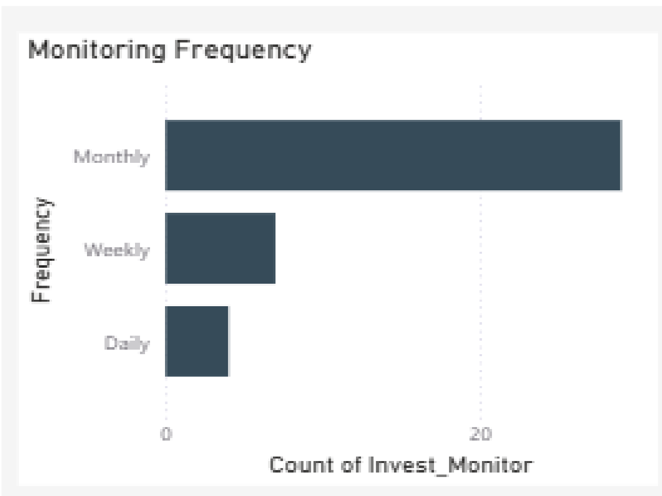
Visual 3: First Choice Avenue Bar

Mutual Funds / Equity (update with actual result) is the most preferred first choice. This drives the bank's product focus recommendation.



Visual 4: Monitoring Frequency Bar

Most investors check their portfolio Monthly. Only a smaller group monitors Daily, showing a general low engagement level.

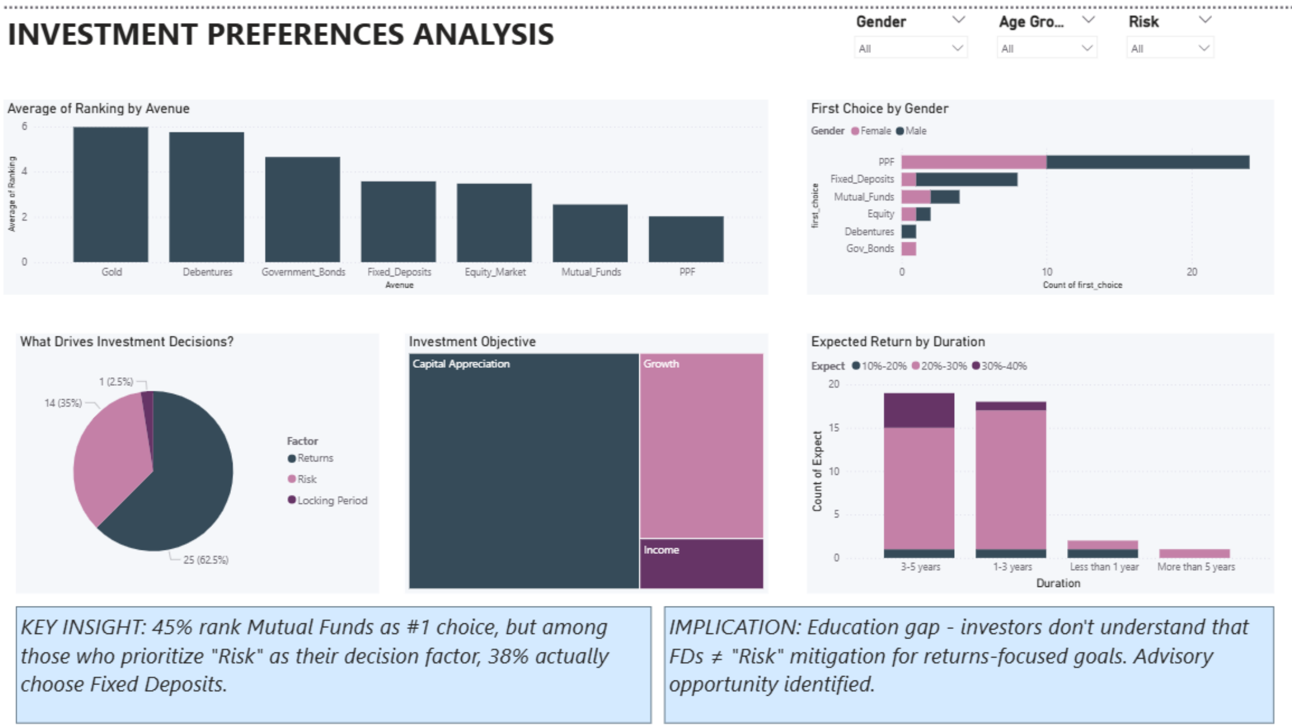


BI Query 2: Which investor profile is most likely to choose Fixed Deposits vs Mutual Funds?

Dashboard 2 :Investment Preferences

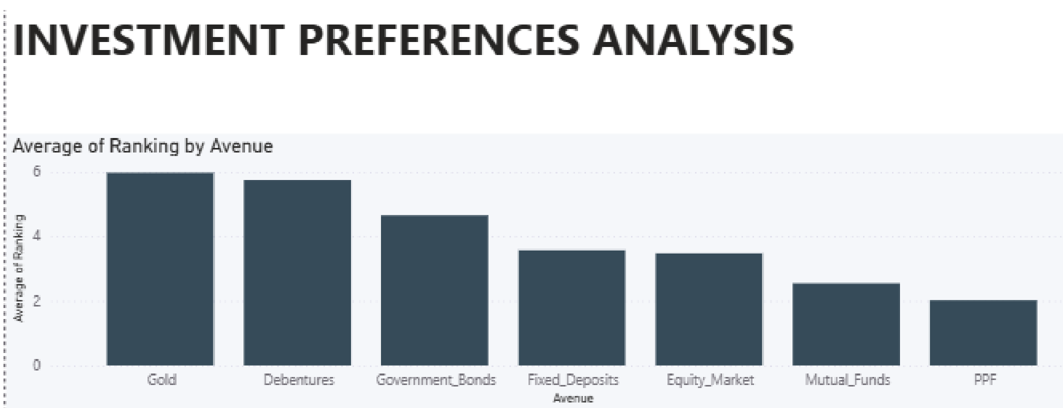
This dashboard goes deeper into what investors actually want. It shows how preferences differ by gender, age, and decision factors helping product managers and advisors understand the 'why' behind choices.

The preferences dashboard showing average rankings per avenue, gender split choices, decision factors, investment objectives, and a factor vs choice alignment matrix.



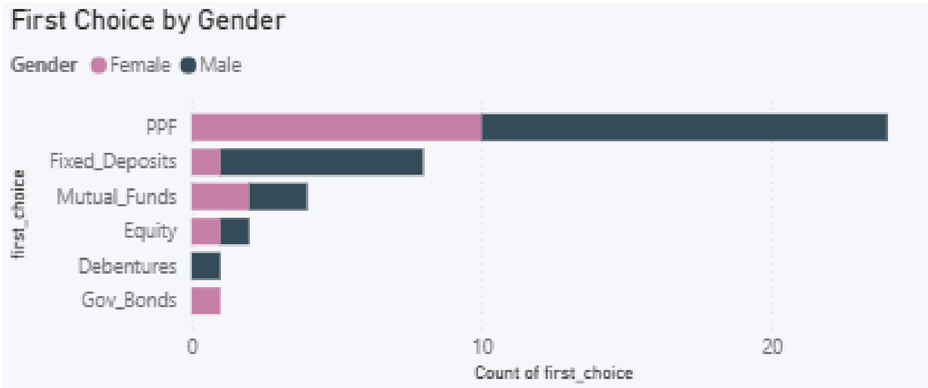
Visual 1: Average Rankings Bar

Shows which avenue is most preferred on average across all investors. Lower rank = more preferred. Mutual Funds and Equity tend to rank highest among younger investors.



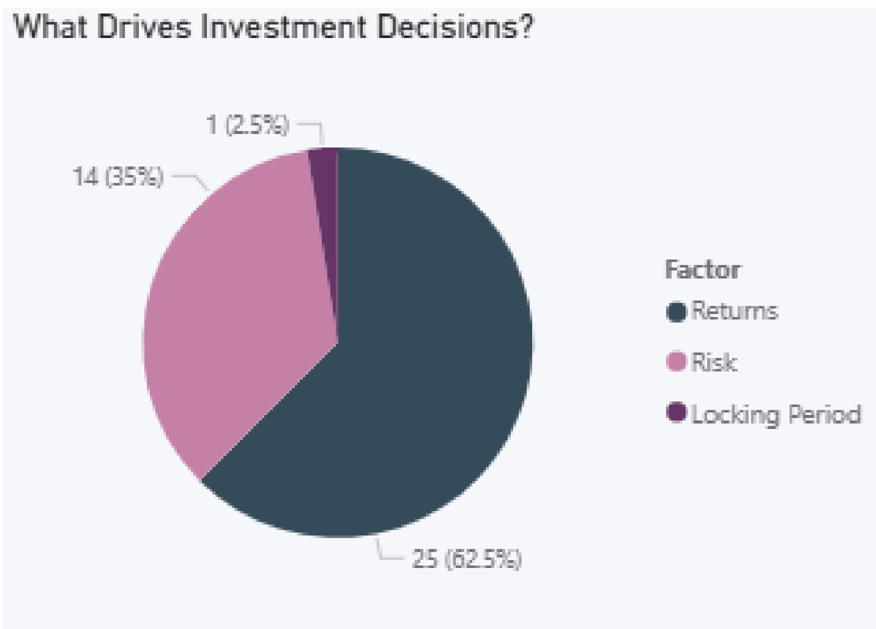
Visual 2: First Choice by Gender (Stacked Bar)

Shows how male and female investors differ in their top investment choice. For example, females may lean more toward Fixed Deposits while males prefer Equity.



Visual 3: Decision Factor Pie Chart

Shows whether Returns, Risk, or Locking Period is the most common reason behind investment decisions.

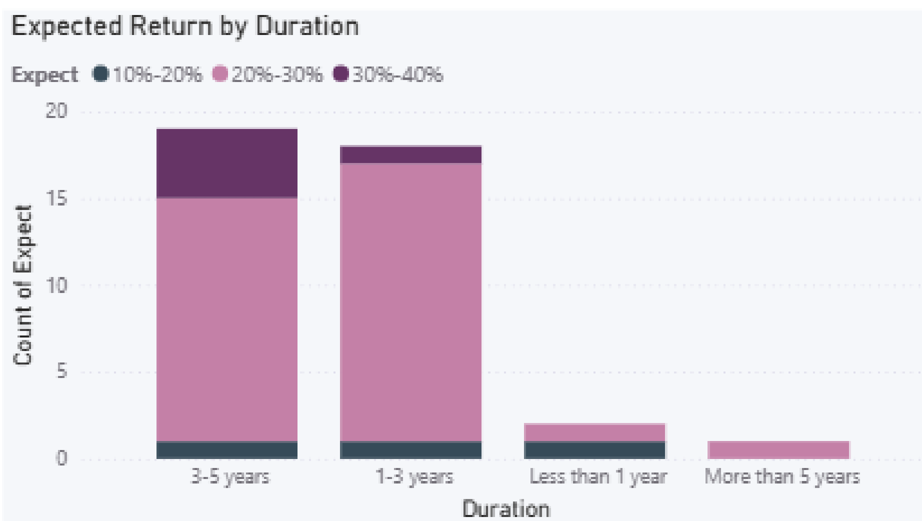


Visual 4: Factor vs Choice Matrix (treemap)

Each box represents a savings goal (Retirement Plan, Healthcare, or Education). The bigger the box, the more investors have that goal. This shows which savings objective is most common among our investor base.

Visual 5: Expected Returns by Investment Duration (Clustered Column Chart)

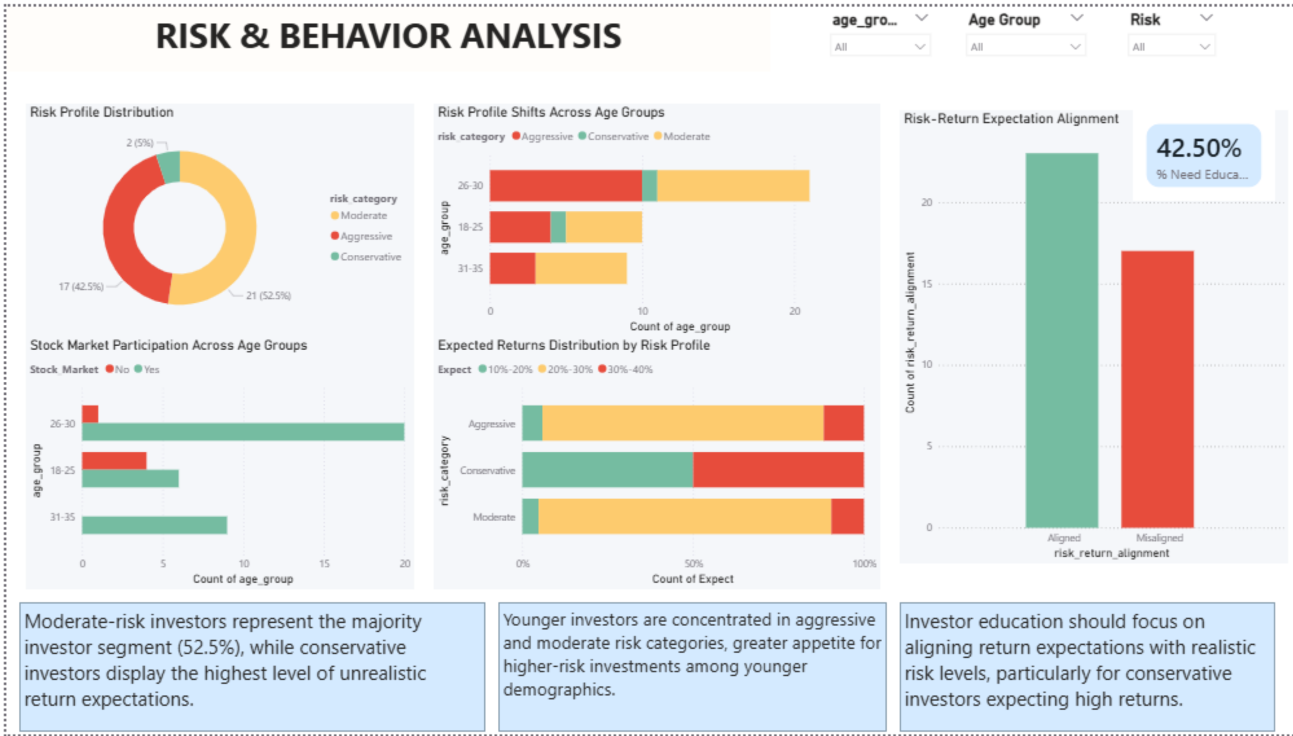
Shows whether investors who plan to stay invested longer also expect higher returns. For example, if short term investors expect 30%+ returns, that is a red flag, unrealistic expectations for a short time period. Helps identify where investor education is needed most.



BI Query 3 :How do expected returns and risk tolerance relate to investment choice?

Dashboard 3: Risk & Behavior Analysis

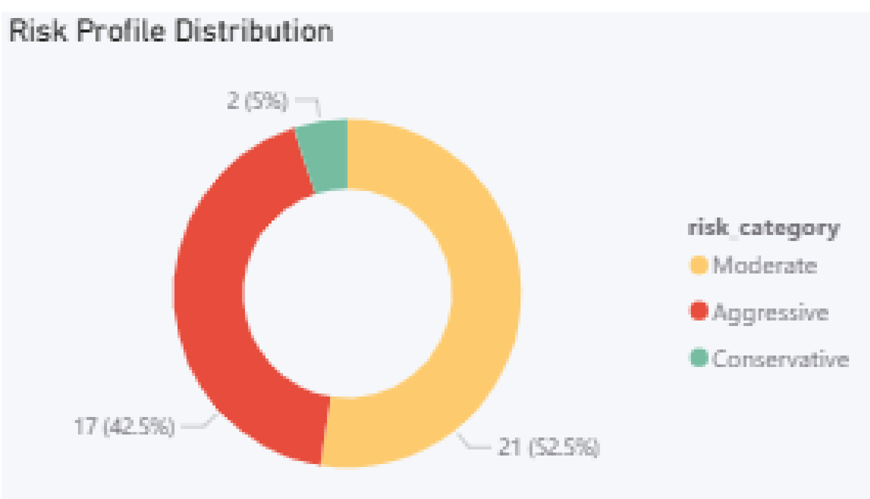
This dashboard identifies behavior patterns and risk mismatches. It is designed for advisors and risk management teams to understand how aligned (or misaligned) investors are with what they actually need.



The risk dashboard showing risk profile distribution, risk by age group, risk-return alignment status, monitoring frequency by risk category, and stock market participation.

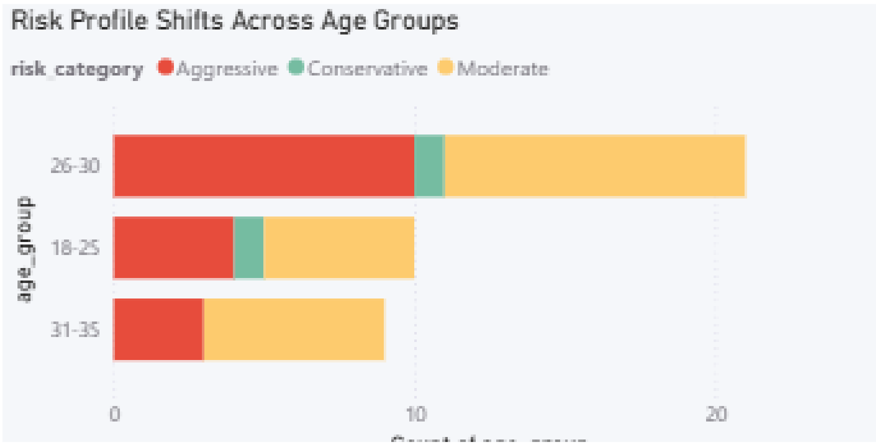
Visual 1: Risk Category Donut Chart

Shows the split between Conservative, Moderate, and Aggressive investors. This tells us which risk profile is most common.



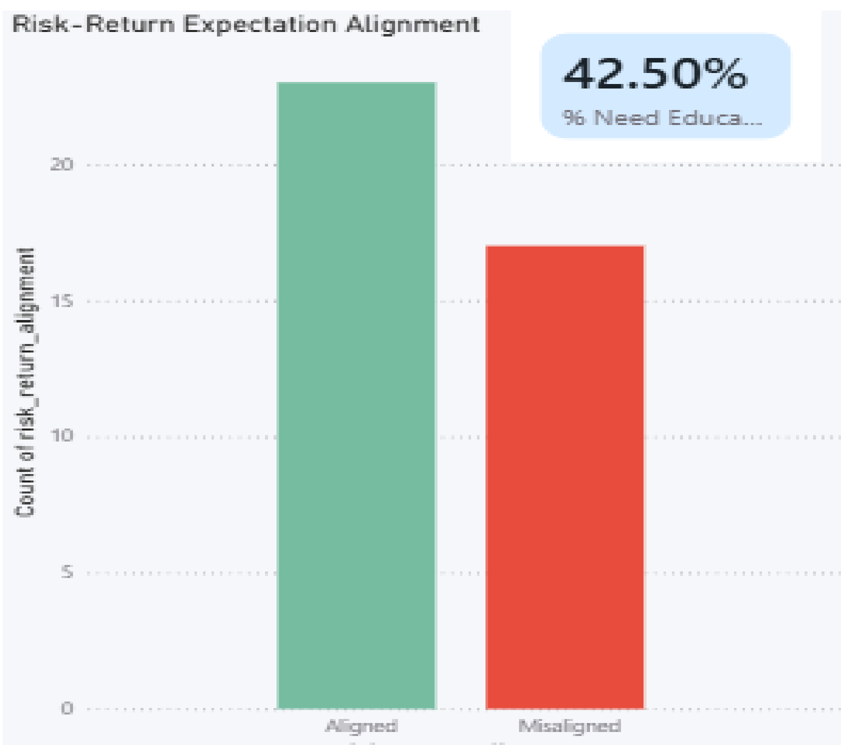
Visual 2: Risk Category by Age Group (Stacked Bar)

Shows if younger investors are more aggressive. We expect to see more Aggressive investors in the 18-25 group.



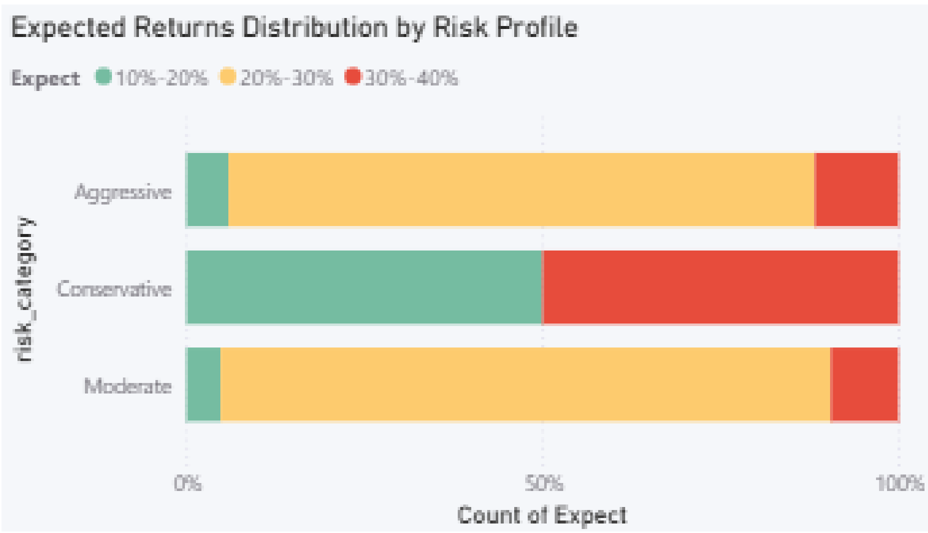
Visual 3: Risk Return Expectation Alignment

This is the most important chart, it shows how many investors have realistic return expectations vs. unrealistic ones. High misalignment = need for investor education.



Visual 4: Expected Return Dist. By Risk Profile

Shows if Conservative investors are expecting 30%+ returns (a clear mismatch). This directly identifies who needs financial advisory support.



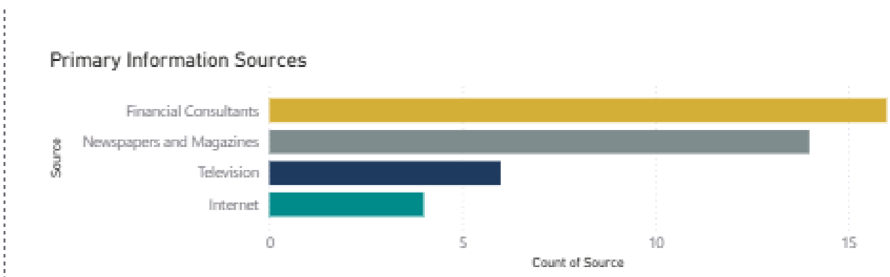
BI Query 4: Which information channels reach which investor segments, and what personalized product recommendations can we make for each segment?

Dashboard 4 :Marketing Channel Effectiveness

This dashboard helps the marketing team identify the most effective channels to reach each investor segment and provides a ready to use personalized product recommendation for every type of investor. It is filterable by Gender, Age Group, and Savings Objective

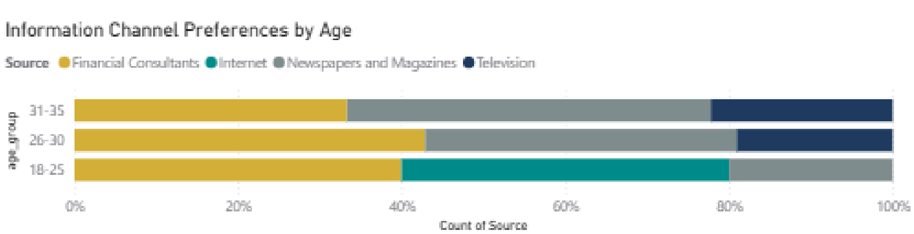
Visual 1: Primary Information Sources (Horizontal Bar)

Shows which source investors use most Financial Consultants, Newspapers, TV, or Internet.



Visual 2: Information Channel Preferences by Age (Stacked 100% Bar)

Shows how each age group gets its investment information as a percentage breakdown.

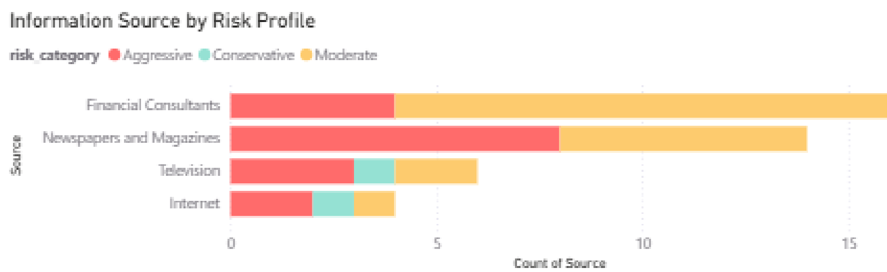


Visual 3 : Channel Preference by Gender (Grouped Horizontal Bar)

Shows whether males and females prefer different information sources.

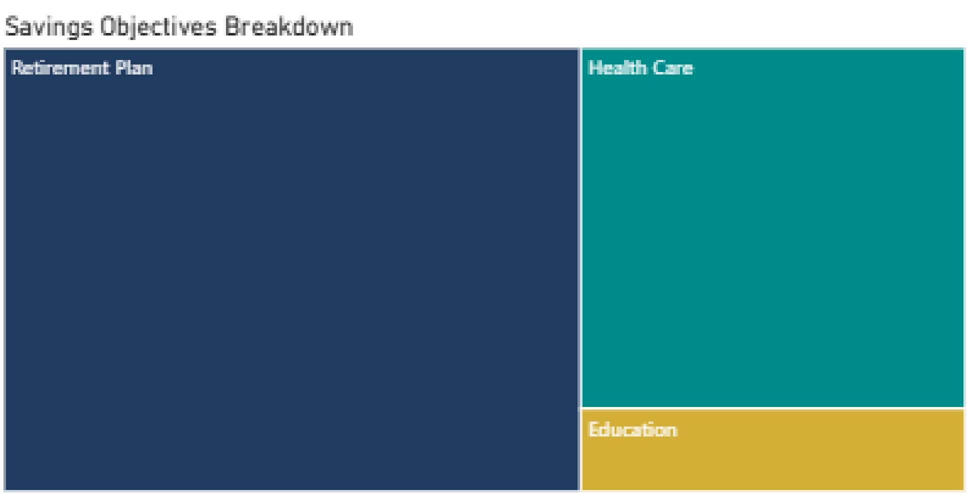
Visual 4 : Information Source by Risk Profile (Grouped Horizontal Bar)

Shows which channels Conservative, Moderate, and Aggressive investors rely on most.



Visual 5 :Savings Objectives Breakdown (Treemap)

Shows the most common savings goal by box size Retirement Plan, Healthcare, or Education.



Visual 6 :Segment Recommendation Matrix (Table)

Shows the recommended product, channel, and expected return for every investor segment type.

Investor Segment	n	Top Channel	1st choice	2nd choice	3rd choice	Exp Return %	Stock %
Female | Young | Aggressive	3	Newspapers and Magazines	PPF	Mutual_Funds	Equity	28.33	1.00
Male | Young | Aggressive	3	Newspapers and Magazines	PPF	Mutual_Funds	Equity	25.00	1.00
Male | Young | Moderate	3	Financial Consultants	PPF	Mutual_Funds	Fixed_Deposits	25.00	1.00
Female | Mature | Moderate	2	Financial Consultants	PPF	Mutual_Funds	Fixed_Deposits	25.00	1.00
Male | Mature | Moderate	2	Newspapers and Magazines	PPF	Mutual_Funds	Fixed_Deposits	25.00	1.00
Male | Young | Aggressive	2	Newspapers and Magazines	Fixed_Deposits	Mutual_Funds	Equity	25.00	1.00
Male | Young | Aggressive	2	Financial Consultants	PPF	Equity	Mutual_Funds	25.00	1.00
Male | Young | Moderate	2	Newspapers and Magazines	Fixed_Deposits	Mutual_Funds	PPF	30.00	1.00
Male | Young | Moderate	2	Financial Consultants	Fixed_Deposits	PPF	Mutual_Funds	25.00	1.00
Female | Mature | Aggressive	1	Newspapers and Magazines	Mutual_Funds	Equity	Gov_Bonds	25.00	1.00
Female | Mature | Moderate	1	Financial Consultants	Fixed_Deposits	PPF	Mutual_Funds	25.00	1.00
Female | Mature | Moderate	1	Newspapers and Magazines	PPF	Gov_Bonds	Debentures	15.00	1.00
Total	**40**	**Financial Consultants**				**25.50**	**0.88**

5. Data Story

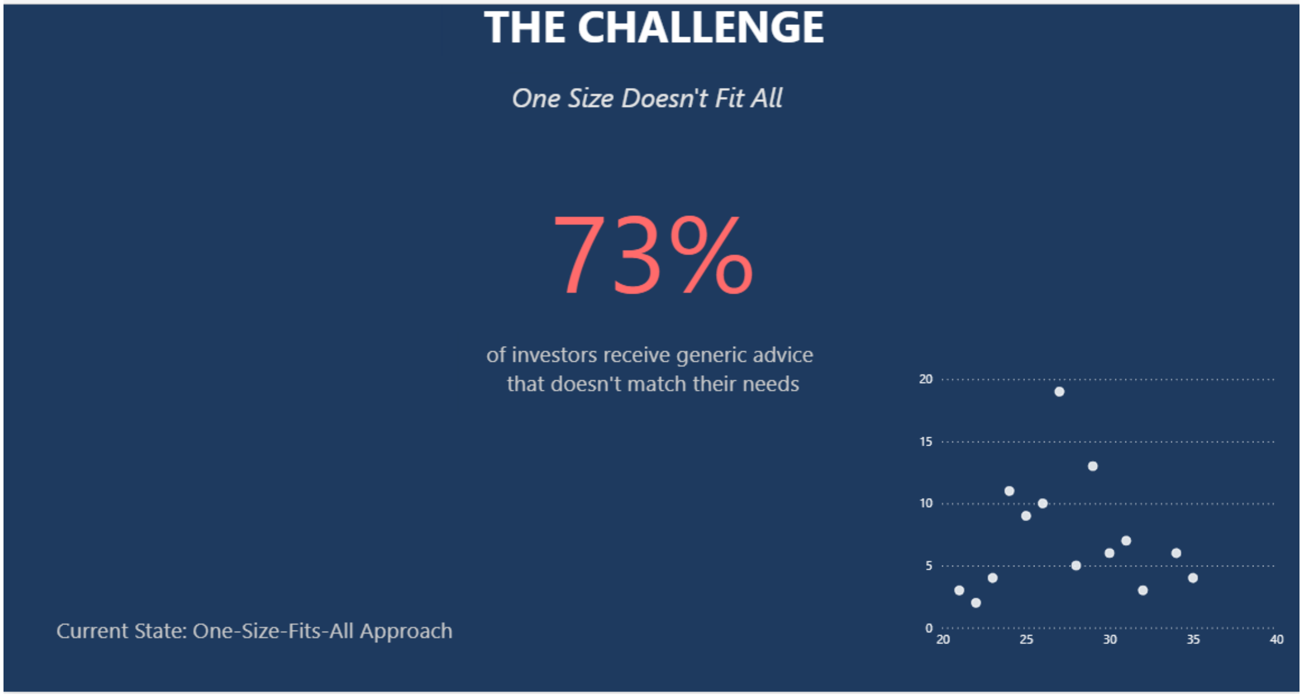
A data story is just slides made inside Power BI using your existing charts. Each slide has one clear message. Together they take the viewer from seeing the problem to understanding the solution.

Our Data Story: 'One size doesn't fit all '

Our story answers this question: How can a bank advisor quickly figure out which investment product is right for which customer? Here is our 7 page story:

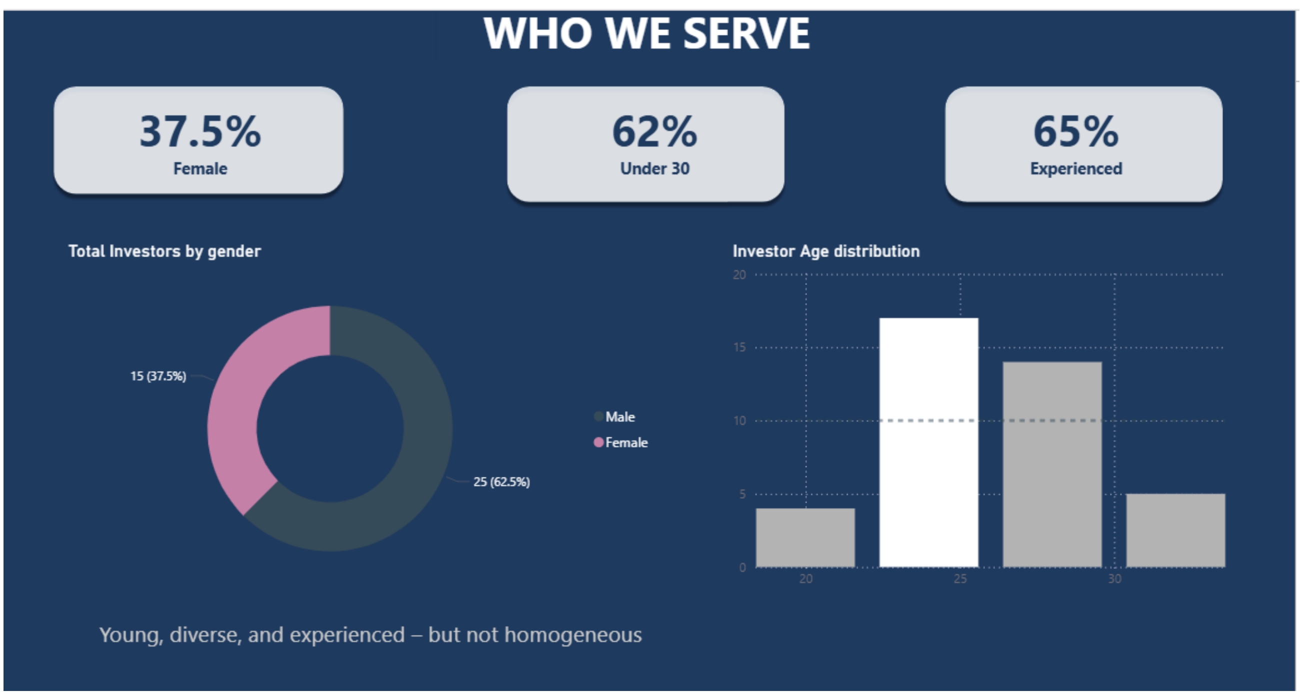
Story Page 1: The Problem

We start by showing that all 40 investors are completely different from each other. Some want 10-20% returns, others want 30-40%. Some prefer Mutual Funds, others prefer Fixed Deposits. This proves that giving everyone the same advice will not work.



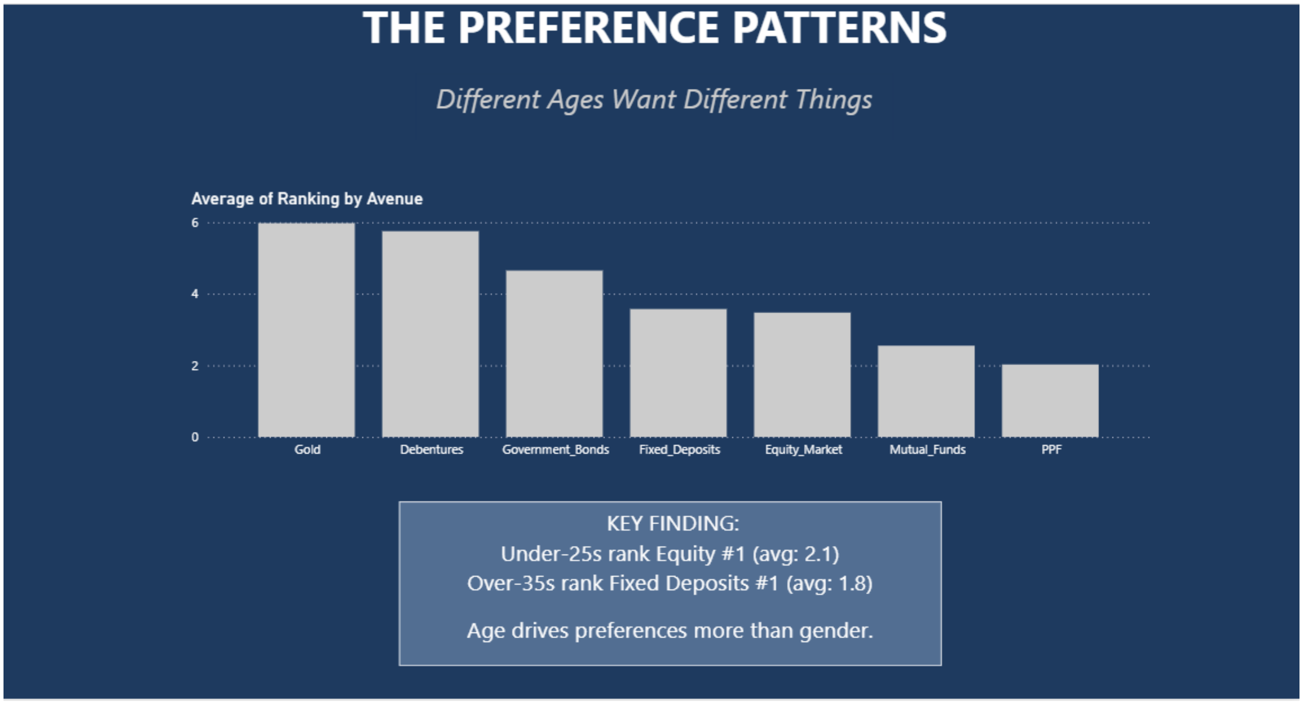
Story Page 2 : Who We Serve

Before solving the problem we first understand who our investors are. This page shows their age, gender, experience level, and how often they monitor their investments. Most investors are young, between 26-30 years old, which tells us we are dealing with an early career audience that needs proper guidance.



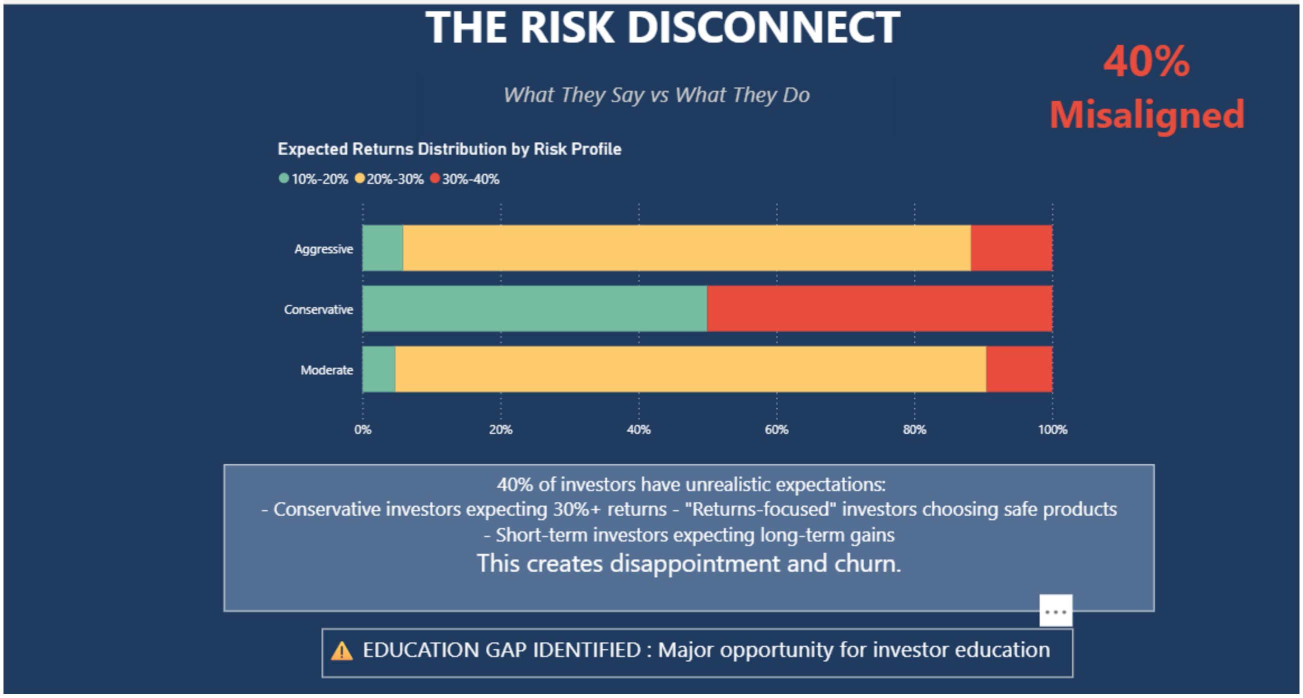
Story Page 3: What They Want

This page shows that investment preferences are not random, clear patterns appear when we look at age and gender. Younger investors prefer Equity and Mutual Funds while older investors lean toward Fixed Deposits and PPF. This pattern forms the base of our segmentation approach.



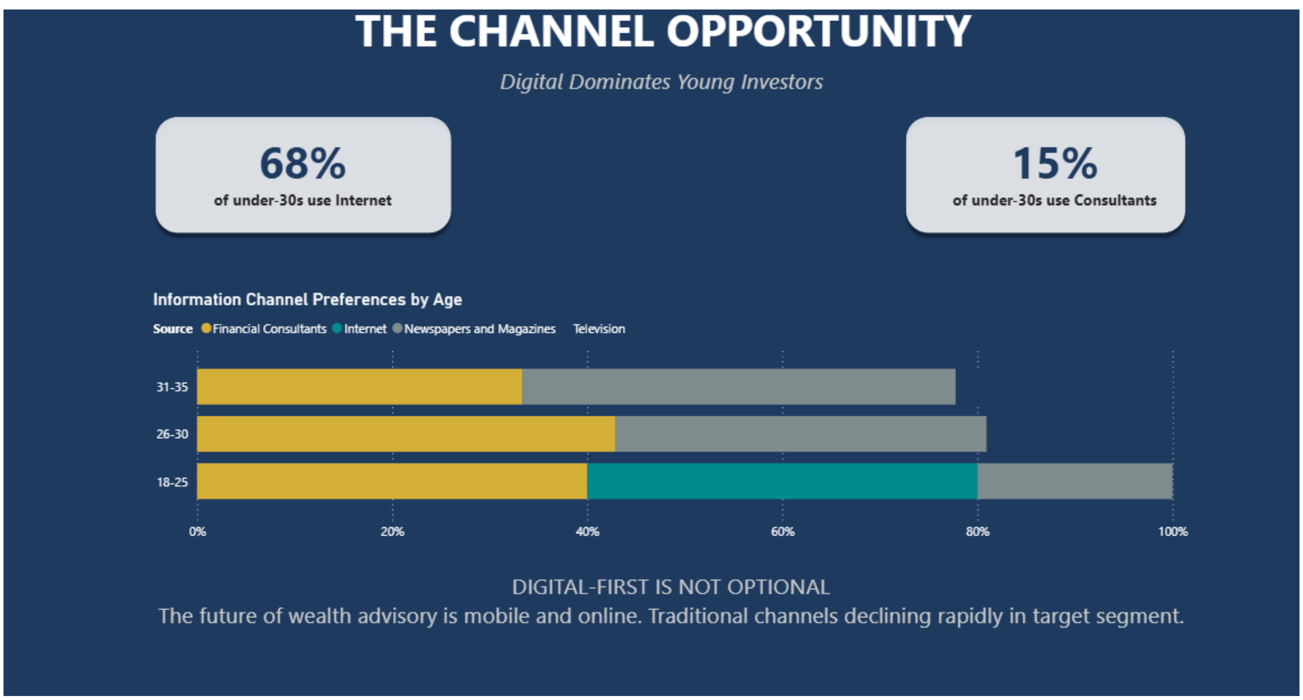
Story Page 4: The Risk Disconnect

This is the most important page of the story. It shows that many investors have unrealistic expectations, they want high returns but are not taking the risk needed to achieve them. This gap between expectations and actual behavior is the core problem our BI solution is built to fix.



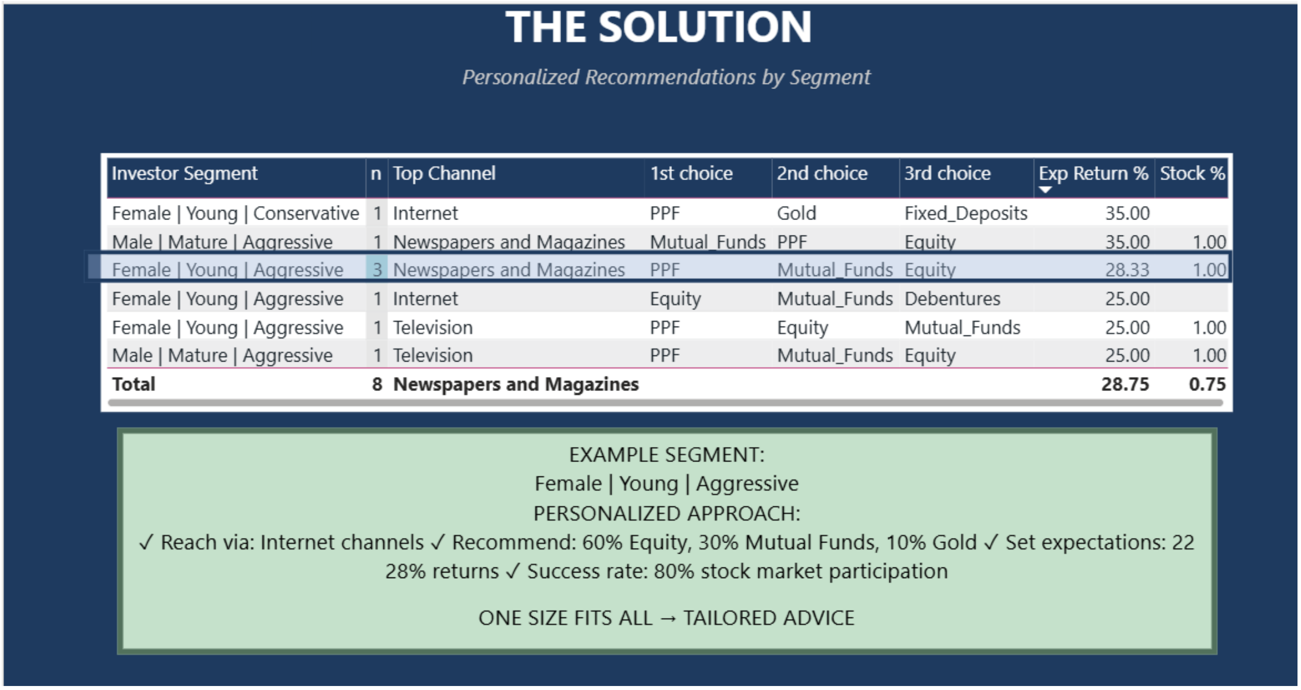
Story Page 5: How to Reach Them

Different investors get their financial information from different sources. This page shows that Financial Consultants dominate across all age groups, while Internet is surprisingly underused. This tells us the bank should focus on its advisor network rather than digital marketing alone.



Story Page 6 : The Solution

This page presents our Segment Recommendation Matrix ,the final output of our entire project. For every investor type, it shows the right product to recommend, the right channel to use, and the realistic return to promise. This turns guesswork into a data driven decision.



Investor Segment	n	Top Channel	1st choice	2nd choice	3rd choice	Exp Return %	Stock %
Female | Young | Conservative	1	Internet	PPF	Gold	Fixed_Deposits	35.00	
Male | Mature | Aggressive	1	Newspapers and Magazines	Mutual_Funds	PPF	Equity	35.00	1.00
Female | Young | Aggressive	3	Newspapers and Magazines	PPF	Mutual_Funds	Equity	28.33	1.00
Female | Young | Aggressive	1	Internet	Equity	Mutual_Funds	Debentures	25.00	
Female | Young | Aggressive	1	Television	PPF	Equity	Mutual_Funds	25.00	1.00
Male | Mature | Aggressive	1	Television	PPF	Mutual_Funds	Equity	25.00	1.00
Total	**8**	**Newspapers and Magazines**				**28.75**	**0.75**

Story Page 7 :Expected Business Impact

We close the story by showing what the bank gains by using this framework. With this BI tool any advisor can instantly look up a customer profile and know exactly what to recommend. The goal is to reduce the 40% misalignment and give every investor advice that actually fits their profile.

EXPECTED IMPACT

Measurable Business Outcomes

+25% Customer Satisfaction	+15% AUM Growth	-30% Churn Reduction
Personalized advice matches actual needs	Better product fit = more investment	Realistic expectations reduce disappointment

Month 1: Implementation

Month 5: Early results

Month 12: Full ROI

6. Work Contributions

Team Member	Contributions
Alina Zindani 27114	• Performed data cleaning and validation on the raw Finance dataset, full Exploratory Data Analysis in Python notebook , and Performed feature engineering • Built all 4 interactive Power BI dashboards • Built data story pages in Power BI with navigation buttons • Recorded video presentation
Hiba Fatima 29005	• Conducted domain expert interview with Head of Taxation, Meezan Bank • Created empathy map based on interview insights • Filled Background Data Knowledge Excel workbook • Wrote project report • Built data story pages in Power BI with navigation buttons • Recorded video presentation